1. Campaign Performance: Retrieve the top 5 campaigns with the highest Click-Through Rate (CTR) over the 30 days period.

```
        SELECT c.campaignName,
        CAST(ROUND((COUNT(clickTimestamp) / COUNT(*)) * 100, 2) AS
FLOAT) AS CTR
        FROM event e
        JOIN campaign c ON e.campaignID = c.campaignID
        JOIN advertiser a ON c.advertiserID = a.id
        WHERE a.advertiserName = %s
        AND timeStamp >= %s
        AND timeStamp < %s
        GROUP BY e.campaignID
        ORDER BY CTR DESC
        LIMIT 5;
```

```
mysql>          SELECT c.campaignName,
    ->          CAST(ROUND((COUNT(clickTimestamp) / COUNT(*)) * 100, 2) AS FLOAT) AS CTR
    ->          FROM event e
    ->          JOIN campaign c ON e.campaignID = c.campaignID
    ->          JOIN advertiser a ON c.advertiserID = a.id
    ->          WHERE a.advertiserName = 'Advertiser_1'
    ->          AND timeStamp >= '2024-10-01 00:00:00'
    ->          AND timeStamp < '2024-11-01 00:00:00'
    ->          GROUP BY e.campaignID
    ->          ORDER BY CTR DESC
    ->          LIMIT 5;
+--------------+------+
| campaignName | CTR  |
+--------------+------+
| Campaign_6   | 5.29 |
| Campaign_4   | 5.26 |
| Campaign_2   | 5.23 |
| Campaign_3   | 5.23 |
| Campaign_5   | 5.15 |
+--------------+------+
5 rows in set (0.56 sec)
```

2. Advertiser Spending: Identify the advertisers that have spent the most money on ad impressions in the last month.

```
        SELECT a.advertiserName, ROUND(SUM(SP_BY_CAMP.cost), 2) AS total
        FROM (SELECT campaignID, SUM(adCost) AS cost
        FROM event
        WHERE timeStamp >= %s
        AND timeStamp < %s
        GROUP BY campaignID
        ) AS SP_BY_CAMP
        JOIN campaign c ON SP_BY_CAMP.campaignID = c.campaignID
        JOIN advertiser a ON a.id = c.advertiserID
        GROUP BY a.advertiserName
        ORDER BY total DESC;
```

```
mysql>         SELECT a.advertiserName, ROUND(SUM(SP_BY_CAMP.cost), 2) AS total
    ->         FROM (SELECT campaignID, SUM(adCost) AS cost
    ->         FROM event
    ->         WHERE timeStamp >= '2024-10-01 00:00:00'
    ->         AND timeStamp < '2024-11-01 00:00:00'
    ->         GROUP BY campaignID
    ->         ) AS SP_BY_CAMP
    ->         JOIN campaign c ON SP_BY_CAMP.campaignID = c.campaignID
    ->         JOIN advertiser a ON a.id = c.advertiserID
    ->         GROUP BY a.advertiserName
    ->         ORDER BY total DESC;
+----------------+-----------+
| advertiserName | total     |
+----------------+-----------+
| Advertiser_19  | 234256.74 |
| Advertiser_99  | 219225.53 |
| Advertiser_12  | 209027.22 |
| Advertiser_59  |  199979.6 |
| Advertiser_87  | 189749.44 |
| Advertiser_39  | 180363.79 |
| Advertiser_95  | 174431.93 |
| Advertiser_11  | 173402.76 |
| Advertiser_66  | 172959.17 |
| Advertiser_52  | 162834.58 |
| Advertiser_42  | 160005.49 |
| Advertiser_34  | 158878.52 |
| Advertiser_4   | 157163.44 |
| Advertiser_33  | 156164.14 |
| Advertiser_67  |  155304.5 |
| Advertiser_30  | 152676.14 |
| Advertiser_49  | 150764.73 |
| Advertiser_57  | 149784.54 |
| Advertiser_45  | 147815.89 |
| Advertiser_36  |  146921.9 |
| Advertiser_91  | 146597.99 |
| Advertiser_71  | 143834.34 |
| Advertiser_9   | 142650.17 |
| Advertiser_54  | 142126.03 |
| Advertiser_46  |  141145.8 |
| Advertiser_86  | 140871.17 |
| Advertiser_25  | 140266.51 |
| Advertiser_92  | 139320.28 |
| Advertiser_6   | 139311.74 |
| Advertiser_48  | 138111.06 |
| Advertiser_78  | 137820.41 |
| Advertiser_58  | 136323.25 |
| Advertiser_65  |  133959.7 |
| Advertiser_26  | 132687.49 |
| Advertiser_97  |  132256.8 |
| Advertiser_21  | 132128.48 |
| Advertiser_73  | 131350.43 |
| Advertiser_85  | 126278.52 |
| Advertiser_1   | 125524.87 |
| Advertiser_60  | 124410.19 |
| Advertiser_8   | 123876.03 |
```

3. Cost Efficiency: Calculate the average Cost Per Click (CPC) and Cost Per Mille (CPM) for each campaign.

```
SELECT c.campaignName, data.CPC, data.CPM
FROM(SELECT campaignID,
ROUND(SUM(adCost) / NULLIF(COUNT(clickTimestamp), 0), 2) AS CPC,
ROUND((SUM(adCost) / (COUNT(*)) * 1000), 2) AS CPM
FROM event
GROUP BY campaignID) AS data
JOIN campaign c ON data.campaignID = c.campaignID
```

```
        ORDER BY data.CPC DESC
```

```
mysql>         SELECT c.campaignName, data.CPC, data.CPM
    ->         FROM(SELECT campaignID,
    ->         ROUND(SUM(adCost) / NULLIF(COUNT(clickTimestamp), 0), 2) AS CPC,
    ->         ROUND((SUM(adCost) / (COUNT(*)) * 1000), 2) AS CPM
    ->         FROM event
    ->         GROUP BY campaignID) AS data
    ->         JOIN campaign c ON data.campaignID = c.campaignID
    ->         ORDER BY data.CPC DESC;
+---------------+-------+---------+
| campaignName  | CPC   | CPM     |
+---------------+-------+---------+
| Campaign_30   | 52.47 | 2285.66 |
| Campaign_998  | 51.38 | 2257.51 |
| Campaign_1    | 51.34 | 2258.07 |
| Campaign_981  | 51.12 |  2295.7 |
| Campaign_69   | 51.03 |  2285.2 |
| Campaign_839  |    51 | 2246.15 |
| Campaign_1009 | 50.85 | 2277.48 |
| Campaign_310  | 50.84 | 2282.51 |
| Campaign_804  | 50.77 | 2278.75 |
| Campaign_634  | 50.67 | 2258.62 |
| Campaign_525  | 50.58 |  2274.9 |
| Campaign_482  | 50.49 | 2273.14 |
| Campaign_882  | 50.34 | 2267.46 |
| Campaign_484  |  50.2 | 2272.28 |
| Campaign_980  |  50.2 | 2259.47 |
| Campaign_381  | 50.14 | 2261.48 |
| Campaign_164  | 50.11 | 2262.48 |
| Campaign_651  | 50.03 | 2284.32 |
| Campaign_232  |    50 | 2278.07 |
| Campaign_821  | 49.99 | 2283.76 |
| Campaign_964  | 49.98 | 2274.68 |
| Campaign_994  | 49.89 | 2272.43 |
| Campaign_570  | 49.69 |  2264.8 |
| Campaign_702  | 49.67 | 2239.02 |
| Campaign_437  | 49.63 | 2278.31 |
| Campaign_421  | 49.57 | 2267.78 |
| Campaign_153  | 49.54 | 2262.27 |
| Campaign_83   |  49.5 | 2258.12 |
| Campaign_603  | 49.49 | 2268.09 |
| Campaign_938  | 49.39 | 2288.79 |
| Campaign_132  | 49.37 | 2264.13 |
| Campaign_59   | 49.32 | 2262.01 |
```

4. Regional Analysis: Find the top-performing locations based on total ad revenue generated from clicks.

```
        SELECT l.location, top.revenue
        FROM(SELECT location_id, ROUND(SUM(adRevenue), 2) AS revenue
        FROM event
        GROUP BY location_id) AS top
        JOIN locations l ON l.id = top.location_id
        ORDER BY top.revenue DESC;
```

```
mysql>         SELECT l.location, top.revenue
    ->         FROM(SELECT location_id, ROUND(SUM(adRevenue), 2) AS revenue
    ->         FROM event
    ->         GROUP BY location_id) AS top
    ->         JOIN locations l ON l.id = top.location_id
    ->         ORDER BY top.revenue DESC;
+-----------+-----------+
| location  | revenue   |
+-----------+-----------+
| USA       | 525391.82 |
| Germany   | 525120.88 |
| UK        | 524162.49 |
| Australia |  523598.1 |
| India     | 523132.49 |
+-----------+-----------+
5 rows in set (1.09 sec)
```

5. User Engagement: Retrieve the top 10 users who have clicked on the most ads.

```
SELECT u.id, u.age, u.gender, l.location,
GROUP_CONCAT(i.interest SEPARATOR ', '),
DATE_FORMAT(u.signup_date, '%Y-%m-%d'),
top.total_click
FROM (SELECT userID, COUNT(*) AS total_click
    FROM event
    GROUP BY userID
    ORDER BY total_click DESC
    LIMIT 10) AS top
JOIN users u ON top.userID = u.id
JOIN user_interests ui ON u.id = ui.user_id
JOIN locations l ON l.id = u.location_id
JOIN interests i ON i.id = ui.interest_id
GROUP BY u.id, u.age, u.gender, l.location, u.signup_date
ORDER BY top.total_click DESC;
```

```
mysql>         SELECT u.id, u.age, u.gender, l.location,
    ->         GROUP_CONCAT(i.interest SEPARATOR ', '),
    ->         DATE_FORMAT(u.signup_date, '%Y-%m-%d'),
    ->         top.total_click
    ->         FROM (SELECT userID, COUNT(*) AS total_click
    ->             FROM event
    ->             GROUP BY userID
    ->             ORDER BY total_click DESC
    ->             LIMIT 10) AS top
    ->         JOIN users u ON top.userID = u.id
    ->         JOIN user_interests ui ON u.id = ui.user_id
    ->         JOIN locations l ON l.id = u.location_id
    ->         JOIN interests i ON i.id = ui.interest_id
    ->         GROUP BY u.id, u.age, u.gender, l.location, u.signup_date
    ->         ORDER BY top.total_click DESC;
+--------+-----+--------+----------+-------------------------------------+-----------------------------------+-------------+
| id     | age | gender | location | GROUP_CONCAT(i.interest SEPARATOR ', ') | DATE_FORMAT(u.signup_date, '%Y-%m-%d') | total_click |
+--------+-----+--------+----------+-------------------------------------+-----------------------------------+-------------+
| 105516 |  42 | Female | India    | Sports, Technology                  | 2021-09-23                          |          35 |
| 399752 |  21 | Female | Germany  | Finance, Health, Education          | 2024-01-20                          |          35 |
|  25733 |  40 | Female | Germany  | Health, Finance                     | 2021-02-21                          |          34 |
| 486831 |  46 | Male   | Australia | Travel                              | 2021-06-20                          |          34 |
| 698841 |  50 | Male   | Germany  | Sports, Health                      | 2023-12-01                          |          34 |
|  85646 |  52 | Female | UK       | Education, Gaming, Fashion          | 2023-12-03                          |          33 |
| 117937 |  63 | Male   | Australia | Technology, Sports, Health, Fashion | 2022-06-04                          |          33 |
| 123617 |  62 | Male   | India    | Gaming, Travel, Health              | 2021-11-26                          |          33 |
| 130249 |  46 | Male   | India    | Fashion, Technology, Finance, Sports | 2022-12-08                          |          33 |
| 147358 |  62 | Female | India    | Sports, Finance                     | 2020-01-06                          |          33 |
+--------+-----+--------+----------+-------------------------------------+-----------------------------------+-------------+
10 rows in set (1.70 sec)
```

6. Budget Consumption: Identify campaigns that have spent more than 80% of their total budget and are close to exhausting their funds.

```
SELECT c.campaignName,
ROUND(((cost.total / c.Budget) * 100), 2) AS percent
FROM (SELECT campaignID,
ROUND(SUM(adCost), 2) AS total
FROM event
GROUP BY campaignID) AS cost
JOIN campaign c ON c.campaignID = cost.campaignID
HAVING percent > 80
ORDER BY percent DESC
```

```
mysql>         SELECT c.campaignName,
    ->         ROUND(((cost.total / c.Budget) * 100), 2) AS percent
    ->         FROM (SELECT campaignID,
    ->         ROUND(SUM(adCost), 2) AS total
    ->         FROM event
    ->         GROUP BY campaignID) AS cost
    ->         JOIN campaign c ON c.campaignID = cost.campaignID
    ->         HAVING percent > 80
    ->         ORDER BY percent DESC;
+---------------+---------+
| campaignName  | percent |
+---------------+---------+
| Campaign_120  |  221.65 |
| Campaign_159  |  216.16 |
| Campaign_357  |  213.91 |
| Campaign_450  |  184.98 |
| Campaign_435  |  184.88 |
| Campaign_611  |  173.89 |
| Campaign_799  |  172.01 |
| Campaign_52   |  162.03 |
| Campaign_499  |  155.05 |
| Campaign_371  |  147.32 |
| Campaign_706  |  143.25 |
| Campaign_78   |  142.37 |
| Campaign_802  |  136.37 |
| Campaign_715  |  135.24 |
| Campaign_227  |  131.78 |
| Campaign_59   |  130.25 |
| Campaign_294  |  127.76 |
| Campaign_969  |  125.04 |
| Campaign_191  |  124.15 |
| Campaign_798  |  118.23 |
| Campaign_674  |  115.67 |
| Campaign_615  |  114.42 |
| Campaign_522  |  108.93 |
| Campaign_779  |  108.34 |
| Campaign_901  |  104.35 |
| Campaign_696  |  103.27 |
| Campaign_668  |  100.63 |
| Campaign_918  |  100.15 |
| Campaign_890  |    99.9 |
| Campaign_302  |   98.49 |
| Campaign_397  |   98.21 |
| Campaign_809  |   94.34 |
| Campaign_633  |   93.92 |
| Campaign_844  |   92.14 |
| Campaign_229  |   91.42 |
| Campaign_92   |   91.09 |
| Campaign_830  |   90.57 |
| Campaign_428  |   90.15 |
| Campaign_538  |   89.25 |
| Campaign_469  |   88.78 |
| Campaign_238  |   87.69 |
| Campaign_370  |   85.29 |
| Campaign_620  |   85.16 |
| Campaign_284  |    83.5 |
| Campaign_1013 |    83.3 |
+---------------+---------+
45 rows in set (1.11 sec)
```

7. Device Performance Comparison: Compare CTR across different device types (mobile, desktop, tablet).

```
SELECT d.device_type, data.pr
FROM(SELECT device_id,
COUNT(clickTimestamp) * 100.0 / COUNT(*) AS pr
FROM event
GROUP BY device_id) AS data
JOIN device d ON data.device_id = d.id;
```

```
mysql>          SELECT d.device_type, data.pr
    ->          FROM(SELECT device_id,
    ->          COUNT(clickTimestamp) * 100.0 / COUNT(*) AS pr
    ->          FROM event
    ->          GROUP BY device_id) AS data
    ->          JOIN device d ON data.device_id = d.id;
+-------------+---------+
| device_type | pr      |
+-------------+---------+
| Mobile      | 5.01158 |
| Desktop     | 4.98984 |
| Tablet      | 4.99055 |
+-------------+---------+
3 rows in set (1.12 sec)
```